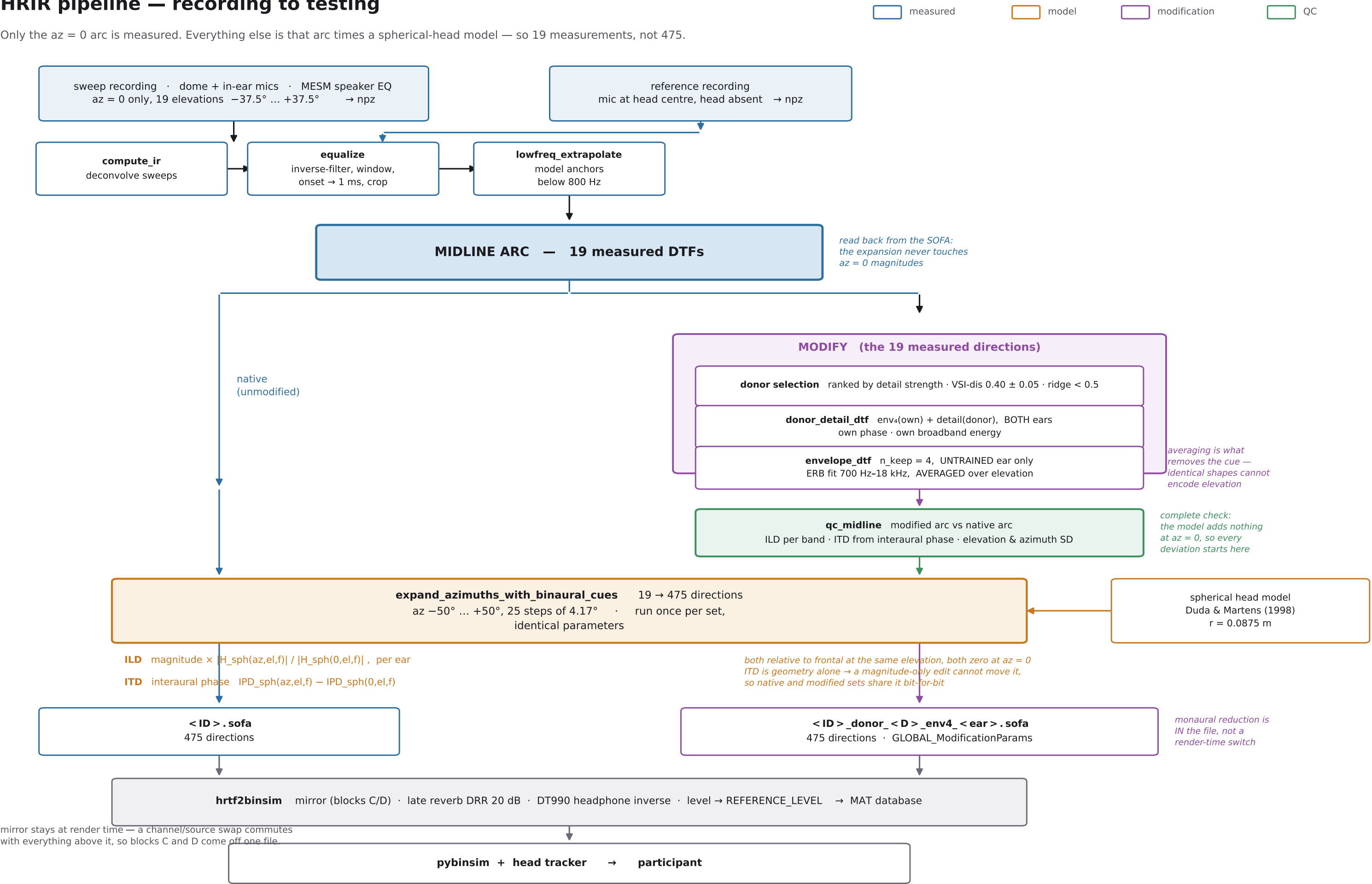


HRIR pipeline — recording to testing

Only the az = 0 arc is measured. Everything else is that arc times a spherical-head model — so 19 measurements, not 475.



Protocol, verification, and build provenance

PROTOCOL

Day 1

· status

· native reference — binaural, native HRTF, full field $\pm 35^\circ$

· build donor (+ pre-stage ranks 1–2 for an in-session swap)

· baseline A — trained ear, modified, trained hemifield

· baseline D — trained ear, MIRRORED modified, mirrored hemifield

Adaptation days (×N)

· PRE test

· train — AR game, 90 s games of head-tracked target trials

· POST test

Final day

· counterbalanced 2×2 (Ear × Side), Williams Latin square

· A trained/same B trained/mirrored

· C untrained/same D untrained/mirrored ← main

localization test — sector (7°, 14°), el $\pm 35^\circ$, 3 targets/sector, min separation 20°, noise, midline excluded for one-sided blocks

BUILD PROVENANCE

v1 — FS, GS, IR, TS, PF. Donor detail on the finished 475-direction SOFA; monaural reduction applied again at render time by hrtf2binsim, per direction, on an already-expanded set. Measured after the fact:

· the untrained ear kept ~half the elevation cue (2.30 dB against 4.66 dB unprocessed, 5.7–11.3 kHz)

· its 0.2–2 kHz ILD sat 9–12 dB from native, further lateral than any real direction in that subject's own HRTF

v2 — from 2026-08. Both steps on the 19 measured az = 0 DTFs, then one expansion. Envelope fitted on an ERB axis and averaged over elevation.

v1 subjects are NOT rebuilt: swapping builds mid-cohort would put a discontinuity inside their own pre/post comparison.

VERIFIED — v2, vs each subject’s own native midline arc

subj	ILD bb	ILD 0.2-2k	ILD 2-16k	ITD	cue leak
FS	0.000	3.12	0.54	0.0000	0.000
GS	0.000	1.47	0.95	0.0000	0.000
IR	0.000	4.03	1.51	0.0000	0.000
TS	0.000	1.04	0.97	0.0000	0.000
PF	0.000	1.46	1.55	0.0000	0.000

dB, dB, dB, μ s, dB. ITD tolerance is 1 μ s — a magnitude-only modification must not move it at all, and it does not.
cue leak = elevation spread of the untrained ear, 5.7–11.3 kHz.

WHAT THE ENVELOPE CHOICE COSTS

	cue leak	ILD 0.2-2k
unprocessed ear	4.66	1.80
linear n=4, per direction (v1)	2.30	10.92
ERB n=4, per direction	1.85	2.10
ERB n=4, elevation-averaged (v2)	0.00	2.22

Smoothing in FREQUENCY has to trade cue removal against level accuracy — both come out of the same n_keep coefficients.
Averaging over ELEVATION removes the cue outright and leaves the level free. Only possible because the arc is all az = 0, so the head shadow is added afterwards by the model and survives.

CONVENTION (unchanged, documented)

SOFA azimuth is CCW-positive: +az is the LEFT hemifield.

TRAINED_HEMI = (−35, 0) for a left-trained subject therefore selects SOFA az 325–360, the RIGHT hemifield — so "trained hemifield" means contralateral to the trained ear. Left as-is for cohort continuity; the 2×2 is invariant under a global sign flip of the azimuth axis.